Macroeconomics I — MPE Insper

Practice list: General Equilibrium, Money and Inflation (Listas 5–6)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — General equilibrium

(Kurlat ch. 9 • 25 points)

Question 1. (10 points) Objective items. The Tesouro asks what pins down the real rate on NTN-Bs in model economies with a representative household and competitive markets.

a) (3 points) Two-period endowment economy, $u = \ln c$, $\beta = 0.96$, $y_1 = 100$, $y_2 = 104$, no government. The equilibrium real interest rate is:

A. 4.00%

B. 8.33%

C. 4.17%

D. 0.00%

b) (3 points) Infinite-horizon economy on a balanced growth path where consumption grows at $g = 2\%$, $u = c^{1-\sigma}/(1-\sigma)$, $\sigma = 2$, $\beta = 0.98$. The real interest rate is:

A. 2.04%

B. 4.08%

C. 8.24%

D. 6.16%

c) (2 points, T or F) In the two-period endowment economy of (a), a rise in patience ($\beta \uparrow$) raises equilibrium saving.

d) (2 points, T or F) By the First Welfare Theorem, the steady-state capital stock of the competitive infinite-horizon economy maximizes steady-state consumption (Golden Rule).

Question 2. (15 points) Discursive. Infinite-horizon economy: household with $\sum_t \beta^t c_t^{1-\sigma}/(1-\sigma)$, inelastic labor $L_t = 1$; firm with $Y = AK^{\alpha}L^{1-\alpha}$; capital depreciates at δ; K_0 given. Numbers: $\beta = 0.95$, $\sigma = 2$, $\alpha = 1/3$, $\delta = 0.1$, $A = 1$.

a) (5 points) Write the household's period budget constraint when it owns and rents the capital, and derive its Euler equation. Solve the firm's problem. Show that $1 + r_{t+1} = r^K_{t+1} + 1 - \delta$ and define a competitive equilibrium (allocation, prices, conditions).

Solution:

b) (5 points) Compute the steady state: r, r^K, K, Y, C, w and the saving rate $\delta K/Y$. Why does σ not appear, and why does r depend only on β?

Solution:

c) (5 points) Compute the Golden Rule K_{gr} and C_{gr} and show $K < K_{gr}$. Explain why this is consistent with the First Welfare Theorem and what happens as $\beta \to 1$. Then let A rise permanently to 1.1: by how much do K, Y, C, w and r change in the new steady state?

Solution:

Block 2 — Money and inflation

(Kurlat chs. 10–11 • 25 points)

Question 3. (10 points) Objective items. Banco Central data and money-demand theory.

a) (3 points) The public holds currency equal to 25% of demand deposits and banks hold reserves equal to 15% of deposits. With a monetary base of R$ 200 billion, M1 is:

A. R$ 500 billion

B. R$ 1,333 billion

C. R$ 625 billion

D. R$ 250 billion

b) (3 points) Real money demand is $m = e^{-a\pi}$ with $a = 4$ (Cagan). The steady-state inflation that maximizes revenue from the inflation tax πm is:

A. 25%

B. 40%

C. 4%

D. 100%

c) (2 points, T or F) If money is superneutral, permanently raising money growth from 4% to 8% leaves the nominal interest rate unchanged.

d) (2 points, T or F) With Baumol–Tobin demand $M/P = \sqrt{FY/(2i)}$ (F a real cost per trip), if real income and the nominal interest rate both double, real balances are unchanged.

Question 4. (15 points) Discursive. Brazil in the 1980s. Real money demand is $M/P = \kappa Y e^{-ai}$ with $i = r + \pi$; Y grows at g, money at μ. Numbers: $\kappa = 0.3$, $a = 2.5$, $r = 0.04$, $g = 0.02$ (illustrative).

a) (5 points) Show that in a steady state with constant i, $\pi = \mu - g$. Write seigniorage $\dot{M}/P$ and the inflation tax $\pi M/P$ as shares of Y as functions of μ, and explain their difference. When does the approximation $i = r + \pi$ fail?

Solution:

b) (5 points) Find the money growth rate that maximizes seigniorage, the associated inflation, the maximum revenue (% of GDP) and the share collected as inflation tax.

Solution:

c) (5 points) The government must finance a deficit of 3% of GDP by printing money. Find the steady states numerically and say which one is on the wrong side of the Laffer curve. What inflation does the Friedman rule prescribe here, and why do central banks such as the BCB target positive inflation instead?

Solution: